Floor Planning/Bin Packing

Sensitivity: Internal

Linear Program

maximise $P = 2x + 3.5y$

subject to:

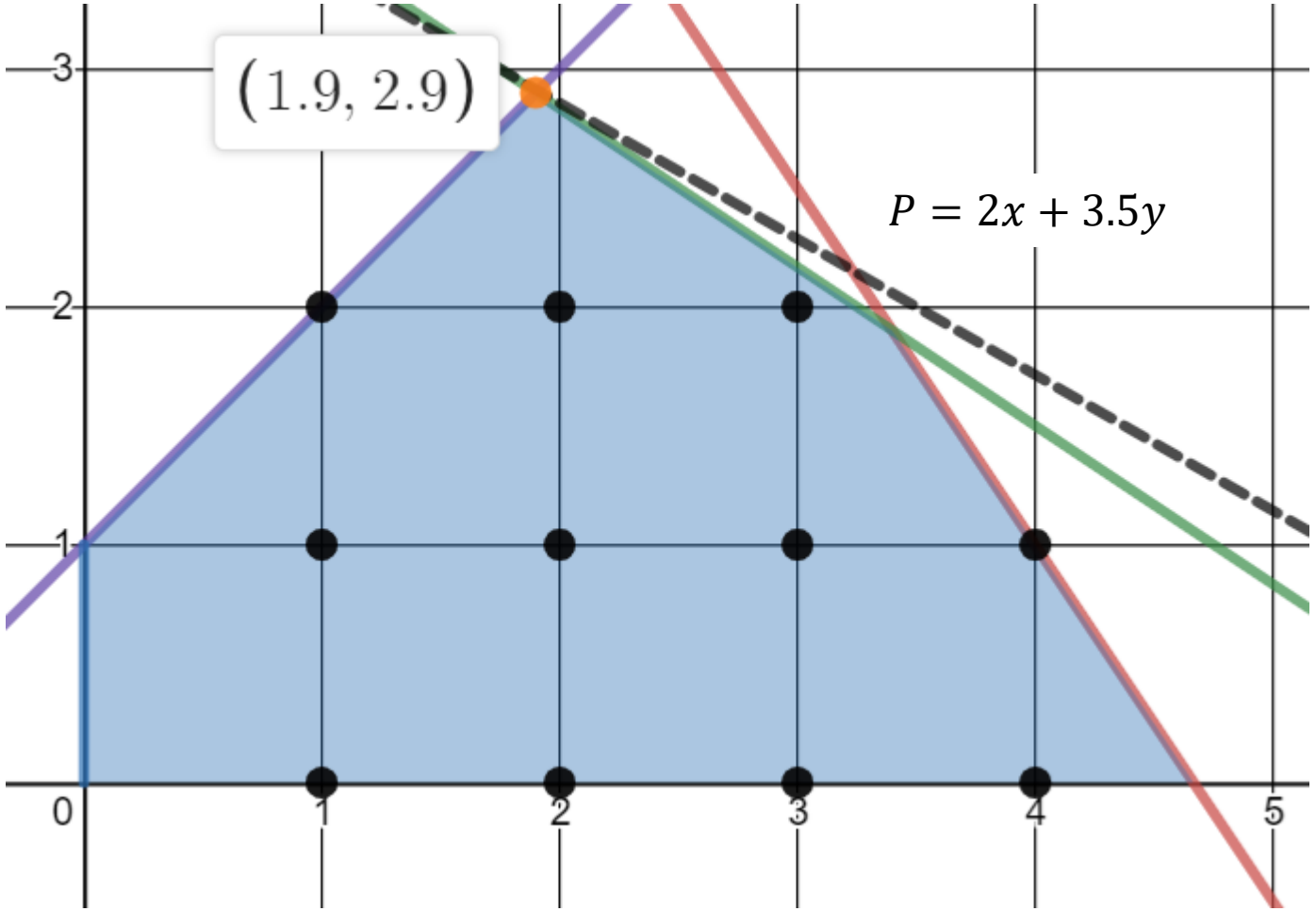
$$3x + 2y \leq 14$$
$$2x + 3y \leq 12.5$$
$$-x + y \leq 1$$

and

$$x, y \geq 0$$

Optimal Solution

$$\boldsymbol{x}^* = (x, y) = (1.9, 2.9)$$
$$P^* = 13.95$$



Sensitivity: Internal

Mixed-Integer Program

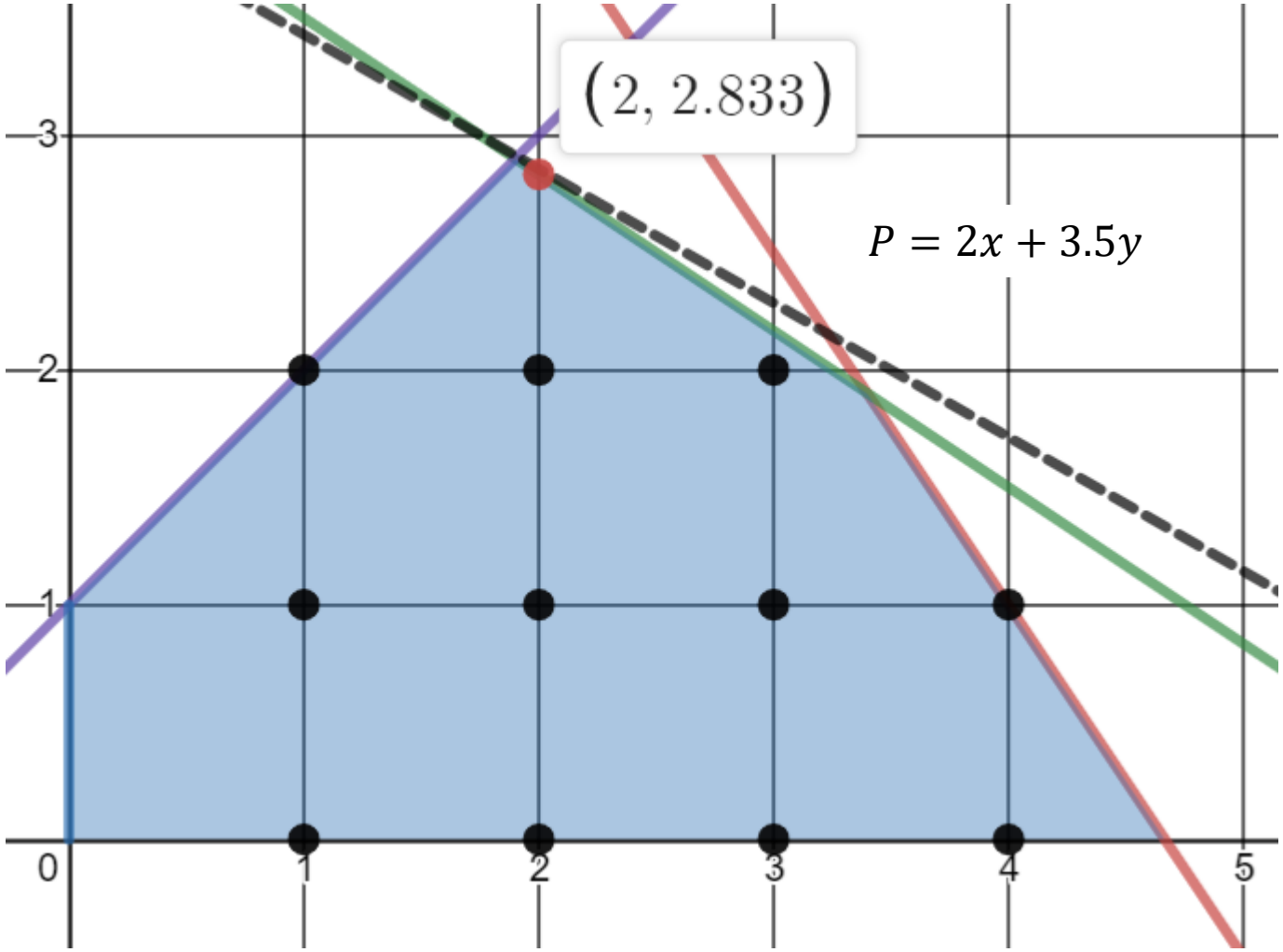
maximise $P = 2x + 3.5y$

subject to:

$$3x + 2y \leq 14$$
$$2x + 3y \leq 12.5$$
$$-x + y \leq 1$$

and

$$x, y \geq 0$$
$$x \in \mathbb{Z}$$



<u>Optimal Solution</u>

$\boldsymbol{x}^* = (x, y) = (2, 2.833)$

$P^* = 13.917$

Sensitivity: Internal

Mixed-Integer Solution (problem based)

maximise $P = 2x + 3.5y$

subject to:

$$3x + 2y \leq 14$$
$$2x + 3y \leq 12.5$$
$$-x + y \leq 1$$

and

$$x, y \geq 0$$
$$x \in \mathbb{Z}$$

Optimal Solution

$\boldsymbol{x}^* = (x, y) = (2, 2.833)$

$P^* = 13.917$

```
1  clear all;
2  % define integer variables
3  x1 = optimvar('x1',1,'LowerBound',0,'UpperBound',inf,'Type','integer');
4  x2 = optimvar('x2',1,'LowerBound',0,'UpperBound',inf);
5  % define objective function values (row vector)
6  f = [2 3.5];
7  % create optimisation problem
8  prob = optimproblem('Objective',sum(f(1)*x1+f(2)*x2),'ObjectiveSense','max');
9  % define the inequality constraints
10 c1 = 3*x1 + 2*x2 <= 14;
11 c2 = 2*x1 + 3*x2 <= 12.5;
12 c3 = -x1 + x2 <= 1;
13 % add them to the optimisation problem
14 prob.Constraints.c1 = c1;
15 prob.Constraints.c2 = c2;
16 prob.Constraints.c3 = c3;
17
18 % convert the optimisation problem
19 problem = prob2struct(prob);
20 % solve and return the solution, function value, exitflag and output
21 [sol,fval,exitflag,output] = intlinprog(problem)
```

```
sol =

    2.0000
    2.8333


fval =

  -13.9167
```

Sensitivity: Internal

2 Dimensional Floor Planning

Sensitivity: Internal

Floor Planning

Pack the following 4 tables into the smallest area possible.

We will build a model which finds the smallest square that holds them all. Even this is complex

$$t_1 - (w_1, h_1) = (4,5)$$
$$t_2 - (w_2, h_2) = (3,7)$$
$$t_3 - (w_3, h_3) = (6,4)$$
$$t_4 - (w_4, h_4) = (7,7)$$

Each t_i has co-ordinates (x_i, y_i) for its position

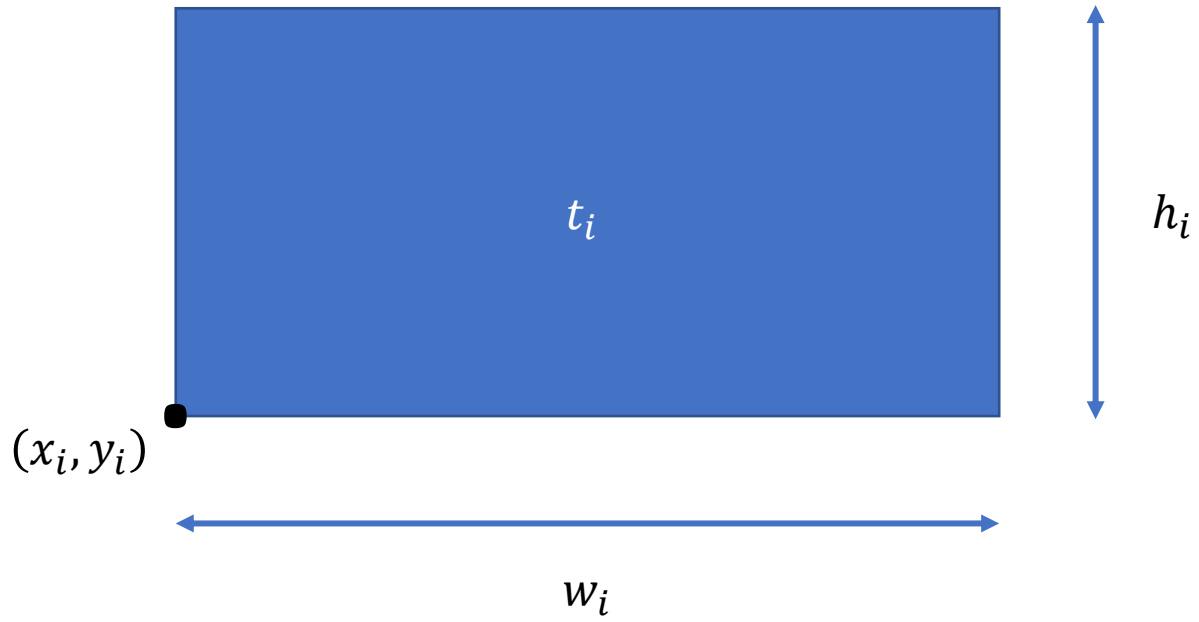
Sensitivity: Internal

Floor Planning

$$t_1 - (w_1, h_1) = (4,5)$$
$$t_2 - (w_2, h_2) = (3,7)$$
$$t_3 - (w_3, h_3) = (6,4)$$
$$t_4 - (w_4, h_4) = (7,7)$$

Each t_i has:

- co-ordinates (x_i, y_i)
- Width - w_i
- Height - h_i



Sensitivity: Internal

Non-overlapping constraints

Make sure that the tables don't overlap.

Consider table t_1 and table t_2

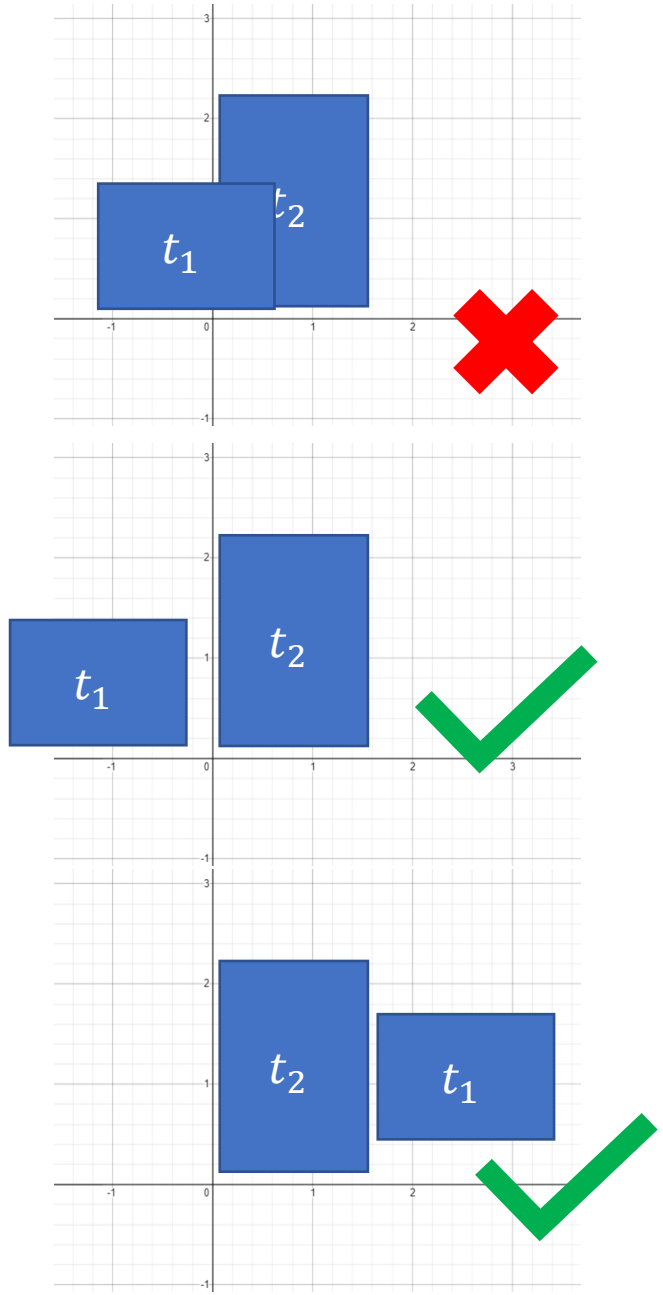
Either

$$x_1 + w_1 \leq x_2$$

Or

$$x_2 + w_2 \leq x_1$$

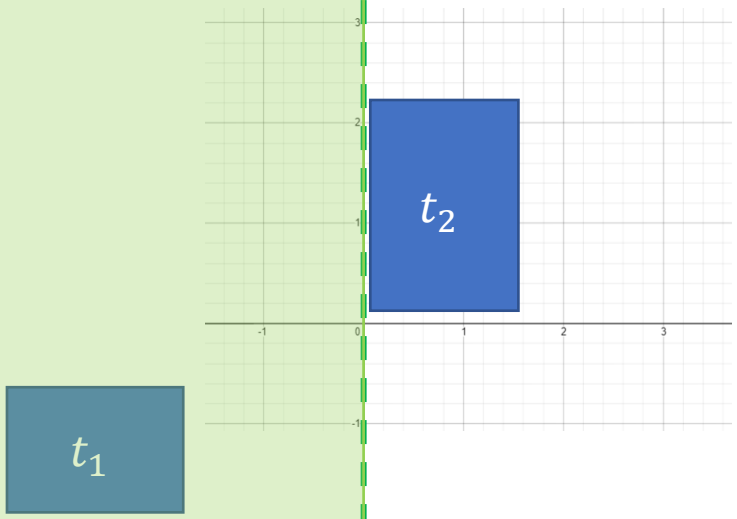
Only one can hold!



Sensitivity: Internal

Intuition – 4 possible situations

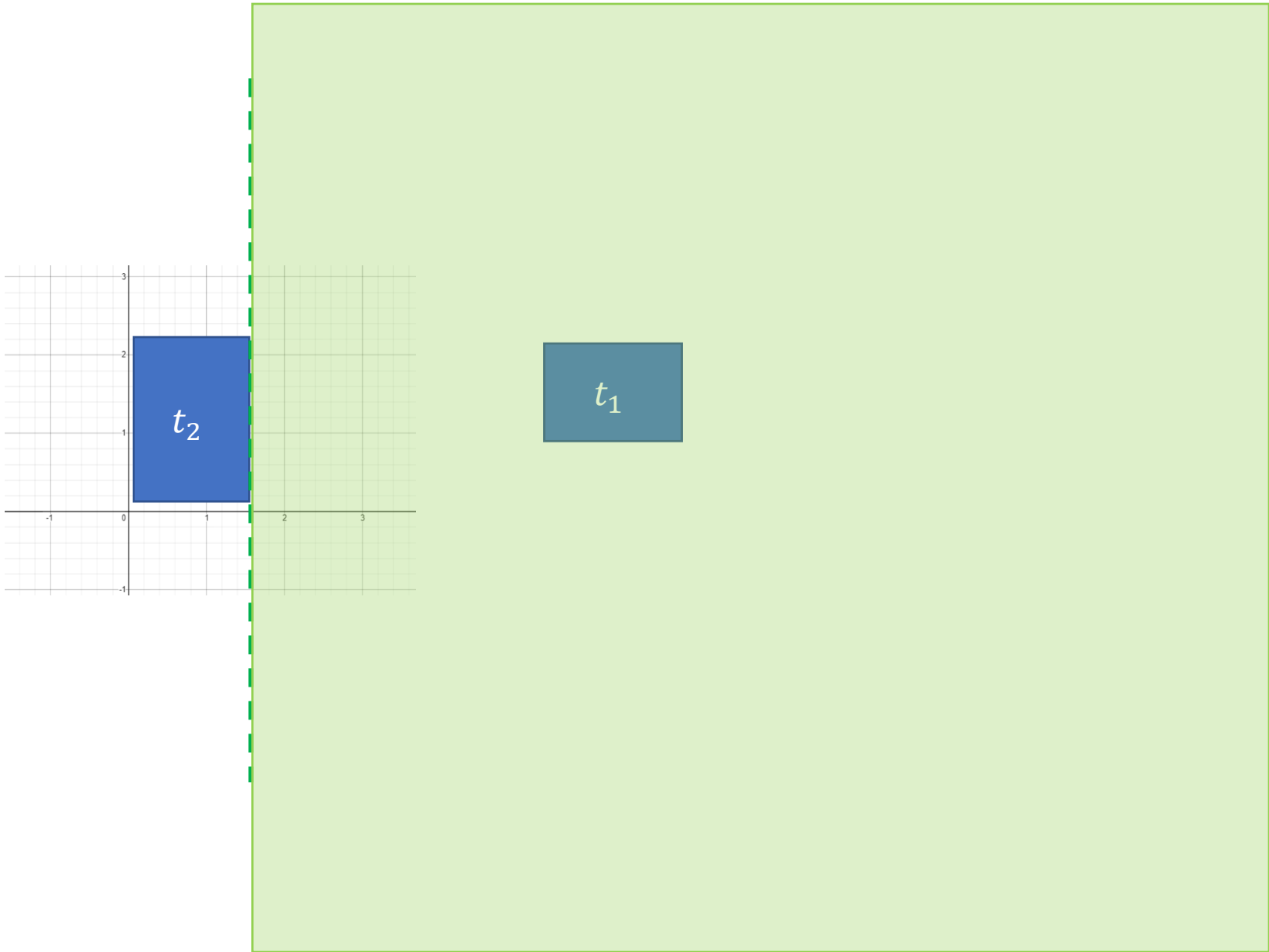
$x_1 + w_1 \leq x_2$



Sensitivity: Internal

Intuition – 4 possible situations

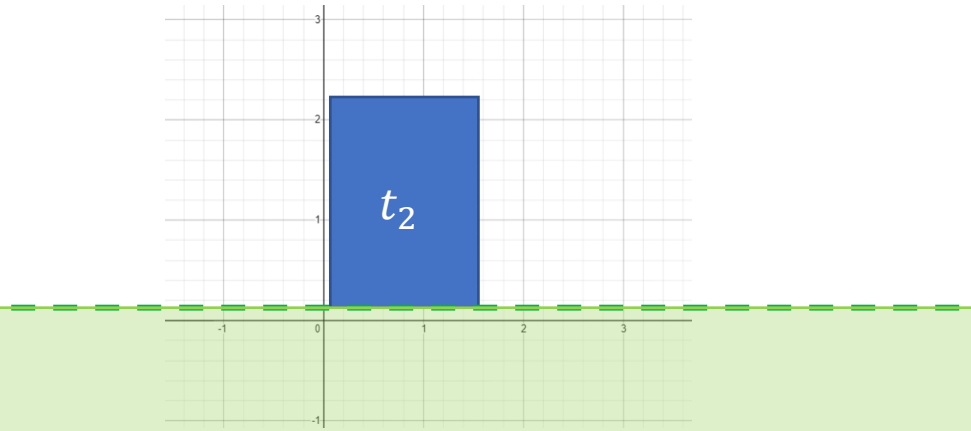
$x_2 + w_2 \leq x_1$



Sensitivity: Internal

Intuition – 4 possible situations

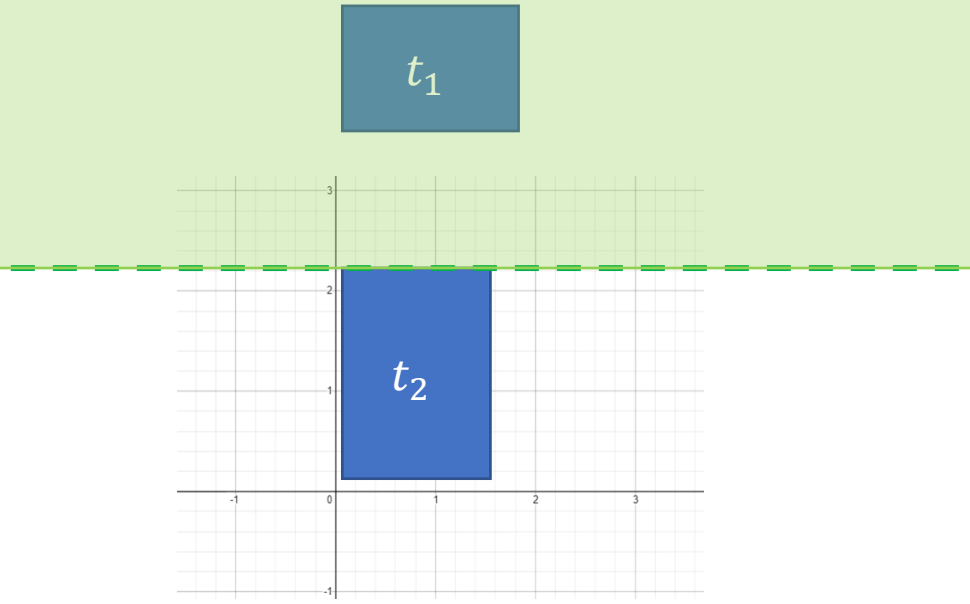
$y_1 + h_1 \leq y_2$



Sensitivity: Internal

Intuition – 4 possible situations

$y_2 + h_2 \leq y_1$



Sensitivity: Internal

Non-overlapping constraints

In fact, two tables don't overlap exactly when at least one of these 4 constraints holds!

Either

$$x_1 + w_1 \leq x_2$$

Or

$$x_2 + w_2 \leq x_1$$

Or

$$y_1 + h_1 \leq y_2$$

Or

$$y_2 + h_2 \leq y_1$$

Sensitivity: Internal

Binary Variables

Introduce binary variables for the two tables

$$x_{12}, y_{12} \in \{0,1\}$$

We can use these to only consider one constraint, let M be a sufficiently large number

x_{12}	y_{12}	$M(x_{12} + y_{12})$	$M(1 - x_{12} + y_{12})$	$M(1 + x_{12} - y_{12})$	$M(2 - x_{12} - y_{12})$
0	0	0	M	M	$2M$
1	0	M	0	$2M$	M
0	1	M	$2M$	0	M
1	1	$2M$	M	M	0

Sensitivity: Internal

Non-overlapping constraints

Our 4 constraints for table 1 and 2 are now

$$x_1 + w_1 \leq x_2 + M(x_{12} + y_{12})$$

$$x_2 + w_2 \leq x_1 + M(1 - x_{12} + y_{12})$$

$$y_1 + h_1 \leq y_2 + M(1 + x_{12} - y_{12})$$

$$y_2 + h_2 \leq y_1 + M(2 - x_{12} - y_{12})$$

x_{12}	y_{12}	C1	C2	C3	C4
0	0	Active	Satisfied	Satisfied	Satisfied
1	0	Satisfied	Active	Satisfied	Satisfied
0	1	Satisfied	Satisfied	Active	Satisfied
1	1	Satisfied	Satisfied	Satisfied	Active

Sensitivity: Internal

Non-overlapping constraints

How many non overlapping constraints?

Test each pair. Given n items (tables)

$$4 \cdot \binom{n}{2}$$

This is a combinatorial problem!

Sensitivity: Internal

Objective

For all i (every table)

$$x_i + w_i \leq A^*$$

$$y_i + h_i \leq A^*$$

Then

$$\text{minimise } A^*$$

This is known as a minimax problem. That is, we wish to minimise the upper bound A^* on the left-hand sides.

Sensitivity: Internal

MILP Formulation

minimise A^*

subject to:

For all i (every table)

$$x_i + w_i \leq A^*$$
$$y_i + h_i \leq A^*$$

MiniMax Constraint

For all $i, j,\ i < j$ (all pairs, don't check twice)

$$x_i + w_i \leq x_j + M(x_{ij} + y_{ij})$$
$$x_j + w_j \leq x_i + M(1 - x_{ij} + y_{ij})$$
$$y_i + h_i \leq y_j + M(1 + x_{ij} - y_{ij})$$
$$y_j + h_j \leq y_i + M\big(2 - x_{ij} - y_{ij}\big)$$

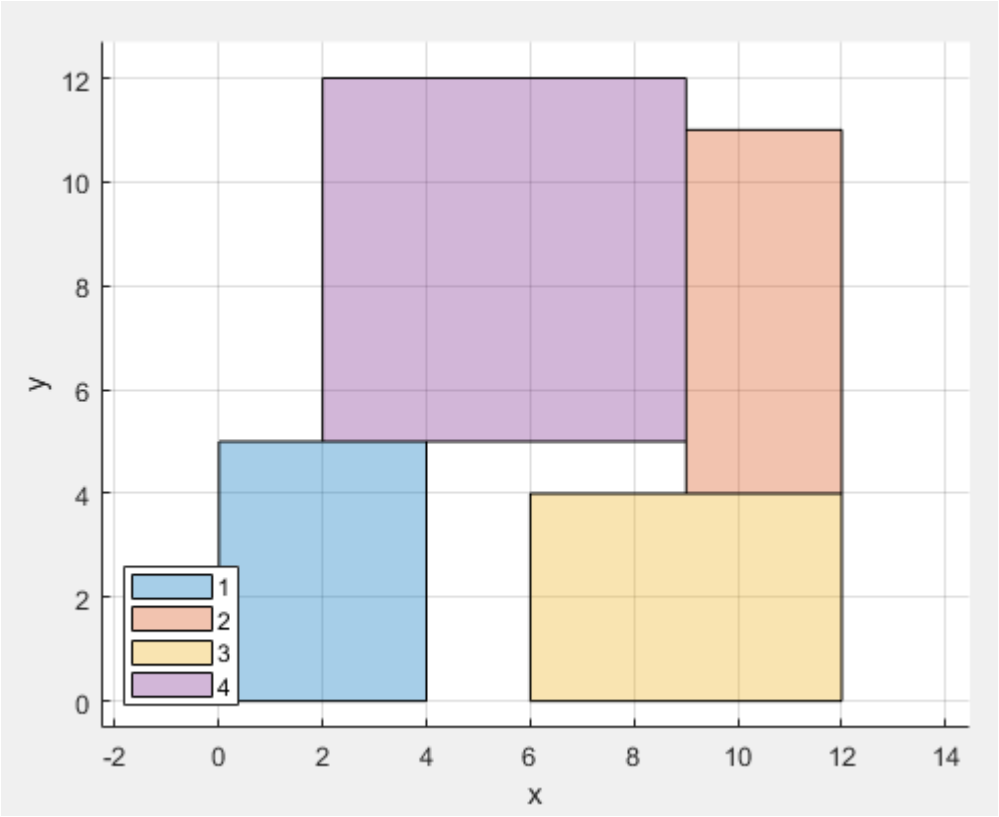
Non overlapping constraints

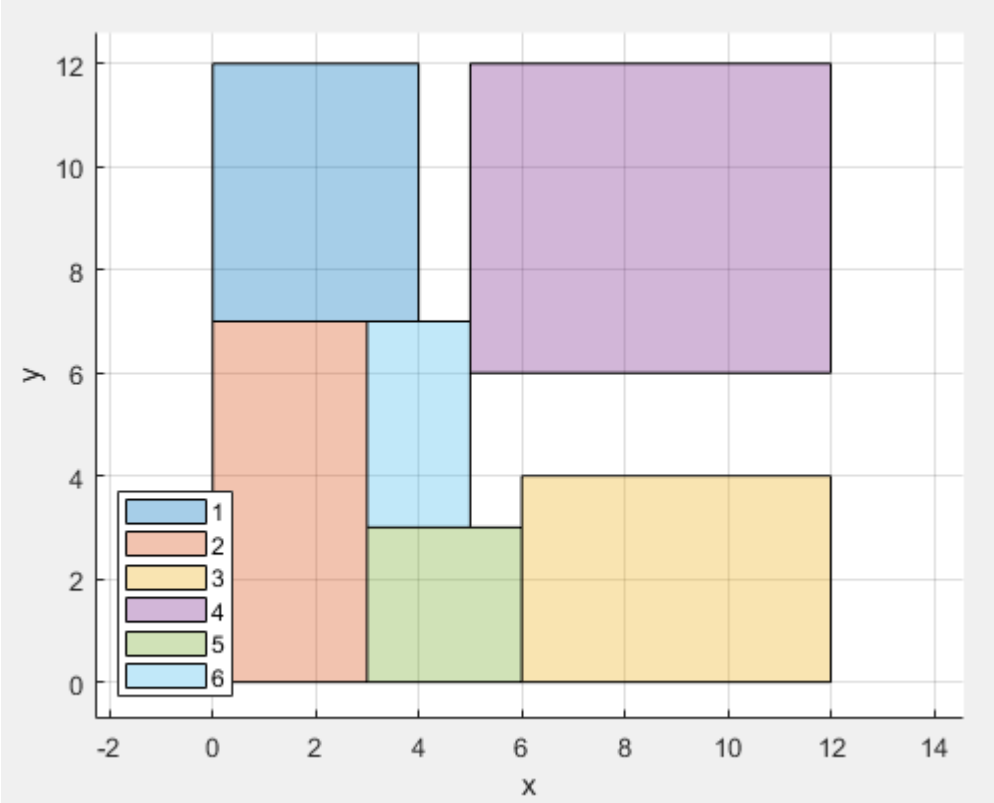
$$x_{ij}, y_{ij} \in \{0,1\}$$
$$x_i, y_i \geq 0$$

Variable constraints

Sensitivity: Internal

Solution





Here is our solution, $A^* = 12$ for both 4 and 6 items!

Our solutions are optimal

Can it be improved? What is the issue?

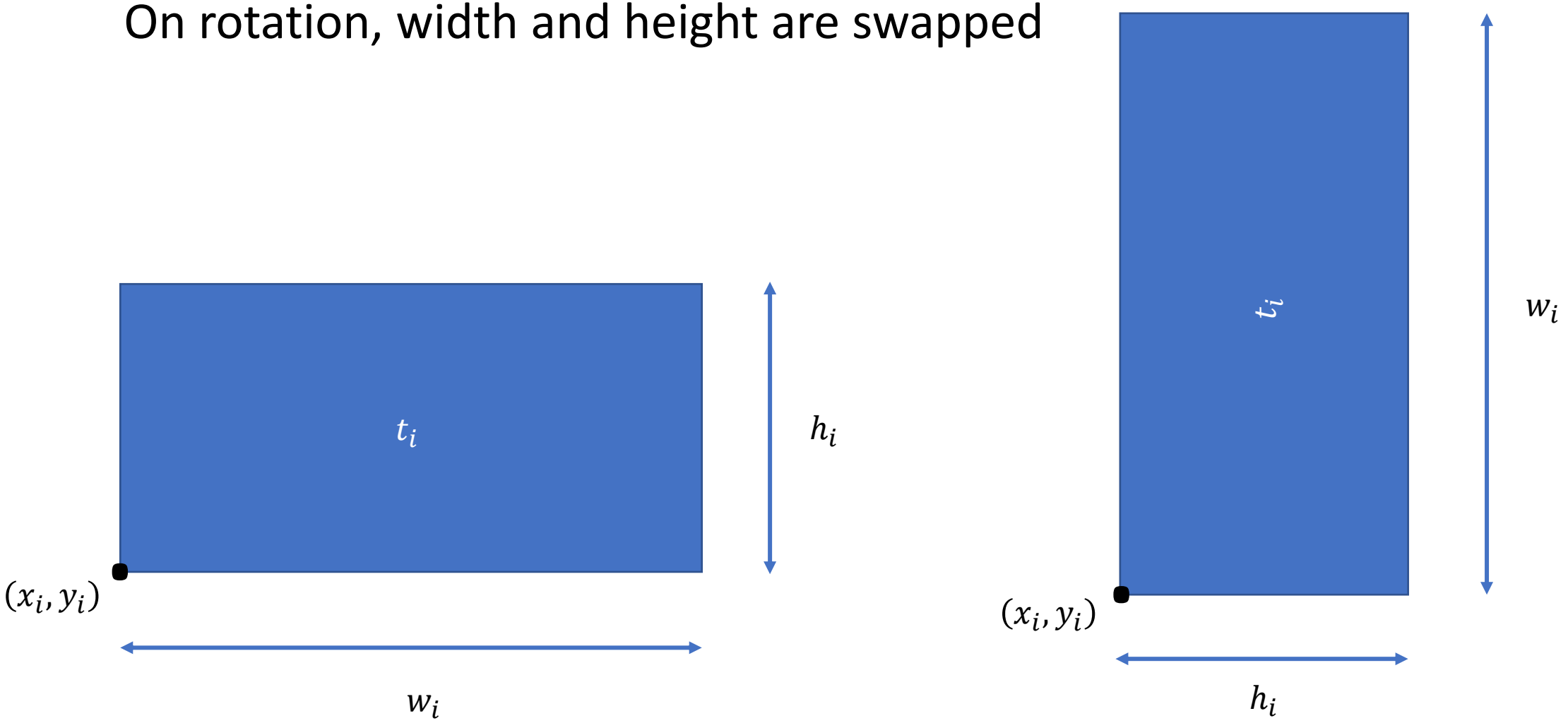
Sensitivity: Internal

Adding Rotation

Sensitivity: Internal

Floor Planning

On rotation, width and height are swapped



Sensitivity: Internal

Amending the constraints

Introduce binary variables r_i. $r_i = 0$ (not rotated), $r_i = 1$ (rotated).

Minimax constraints

For all i (every table)

$$x_i + (1 - r_i)w_i + r_i h_i \leq A^*$$
$$y_i + (1 - r_i)h_i + r_i w_i \leq A^*$$

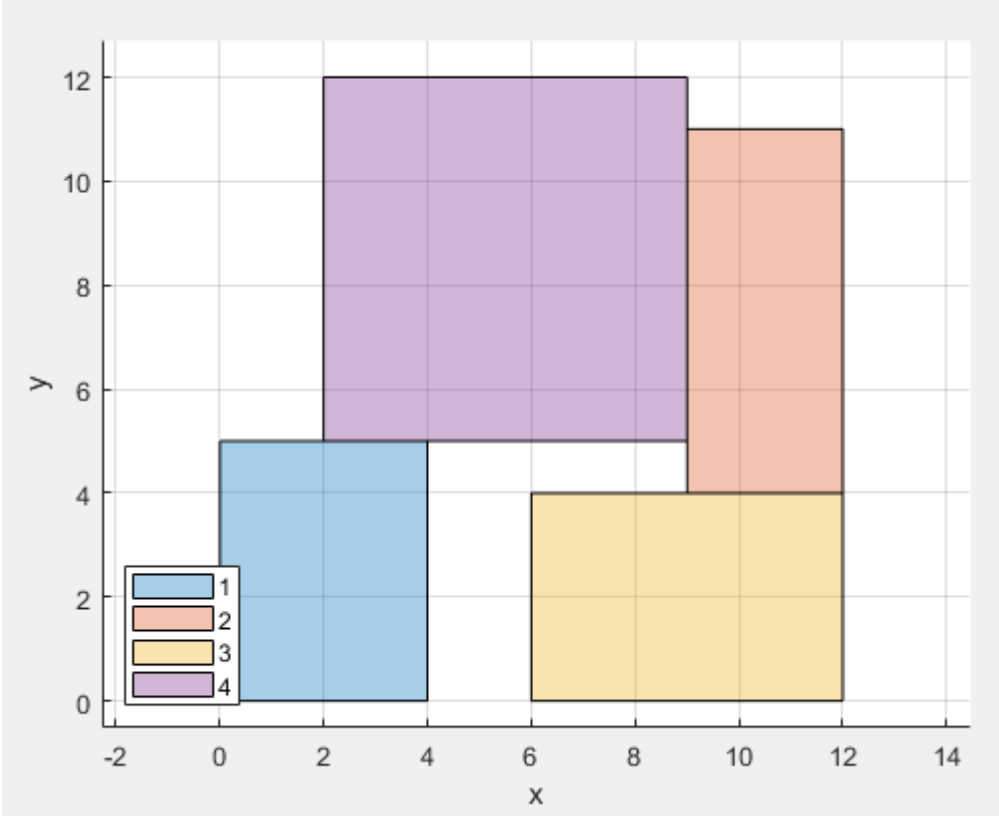
Non overlapping constraints

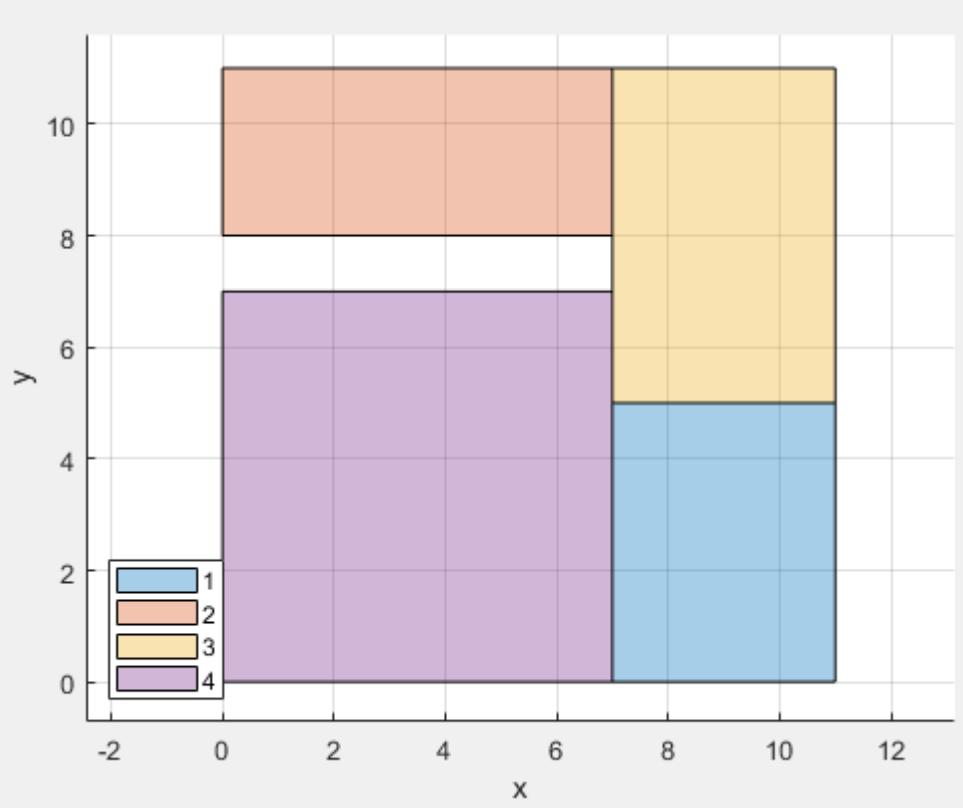
For all $i, j, i < j$ (all pairs, don't check twice)

$$x_i + (1 - r_i)w_i + r_i h_i \leq x_j + M(x_{ij} + y_{ij})$$
$$x_j + (1 - r_j)w_j + r_j h_j \leq x_i + M(1 - x_{ij} + y_{ij})$$
$$y_i + (1 - r_i)h_i + r_i w_i \leq y_j + M(1 + x_{ij} - y_{ij})$$
$$y_j + (1 - r_j)h_j + r_j w_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

Sensitivity: Internal

Solution





Here is our solution, $A^* = 11$. Improvement!

Can it be improved? This is dependent on your desired objective!

Sensitivity: Internal

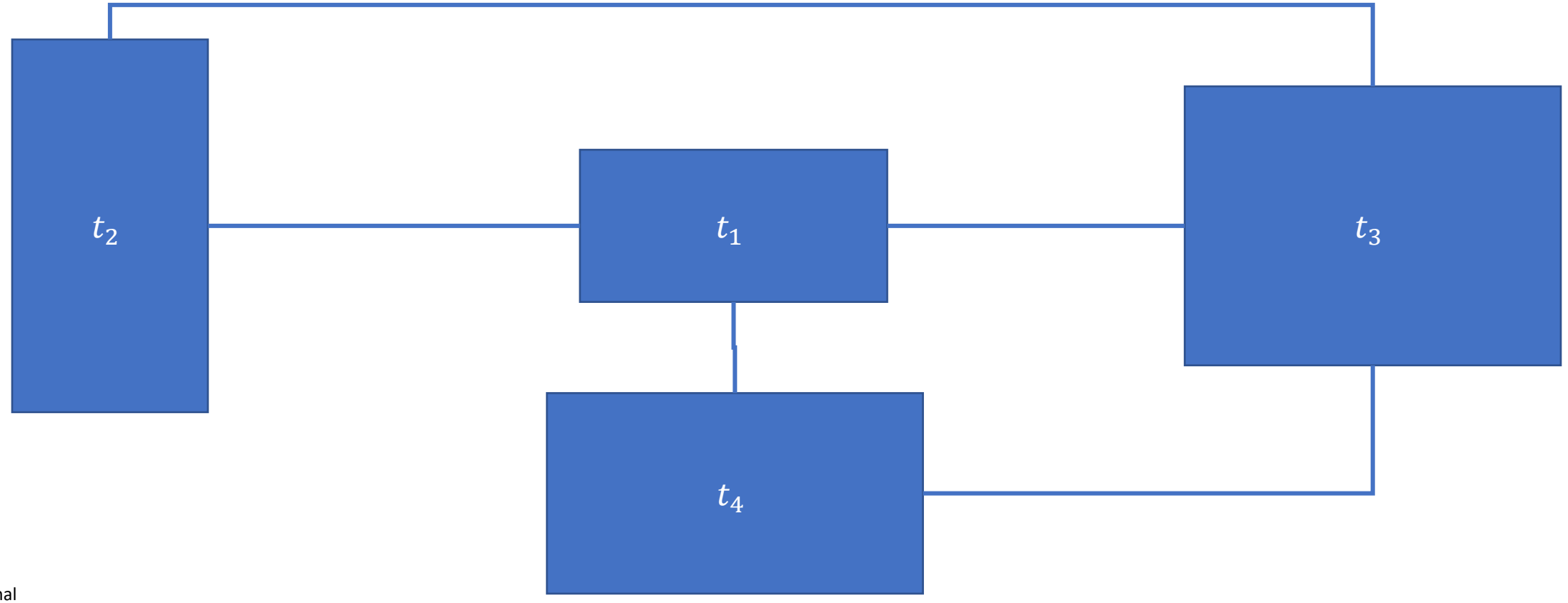
Connected 2 Dimensional Packing

Minimisation of the rectilinear distance between connected items

Sensitivity: Internal

Clustering of components

Perhaps you have connected components that need to be close to each other and you have a fixed width/height for your floor space.



Sensitivity: Internal

Amending the constraints

For a fixed width and height floor.

Boundary constraints

For all i (every table)

$$x_i + (1 - r_i)w_i + r_i h_i \leq W$$
$$y_i + (1 - r_i)h_i + r_i w_i \leq H$$

Note we could drop these if the items did not have to stay within a boundary, or make W, H sufficiently large!

Non overlapping constraints

For all $i, j, i < j$ (all pairs, don't check twice)

$$x_i + (1 - r_i)w_i + r_i h_i \leq x_j + M(x_{ij} + y_{ij})$$
$$x_j + (1 - r_j)w_j + r_j h_j \leq x_i + M(1 - x_{ij} + y_{ij})$$
$$y_i + (1 - r_i)h_i + r_i w_i \leq y_j + M(1 + x_{ij} - y_{ij})$$
$$y_j + (1 - r_j)h_j + r_j w_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

Sensitivity: Internal

Distance between items (from centre points)

For all connected i, j

Let $Dx_{ij} = [x_i + (1 - r_i)\frac{w_i}{2} + r_i\frac{h_i}{2}] - \left[x_j + \left(1 - r_j\right)\frac{w_j}{2} + r_j\frac{h_j}{2}\right]$

We need absolute distance, therefore we force $R_{ij} = \max\{Dx_{ij}, -Dx_{ij}\}$

$$Dx_{ij} \leq \mathrm{R_{ij}}$$
$$-Dx_{ij} \leq \mathrm{R_{ij}}$$

Let $Dy_{ij} = [y_i + (1 - r_i)\frac{h_i}{2} + r_i\frac{w_i}{2}] - \left[y_j + \left(1 - r_j\right)\frac{h_j}{2} + r_j\frac{w_j}{2}\right]$

We need absolute distance, therefore we force $B_{ij} = \max\{Dy_{ij}, -Dy_{ij}\}$

$$Dy_{ij} \leq \mathrm{B_{ij}}$$
$$-Dy_{ij} \leq \mathrm{B_{ij}}$$

Sensitivity: Internal

Objective Function

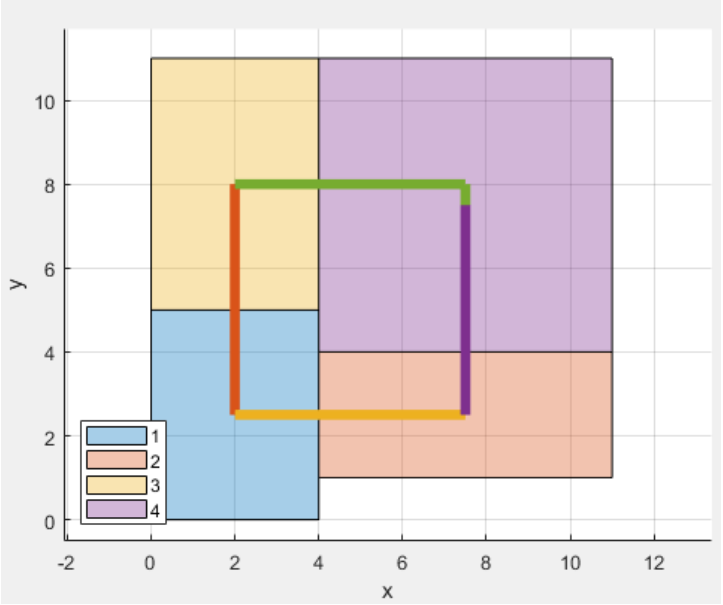
Minimise Rectilinear Distance

For all connected i, j

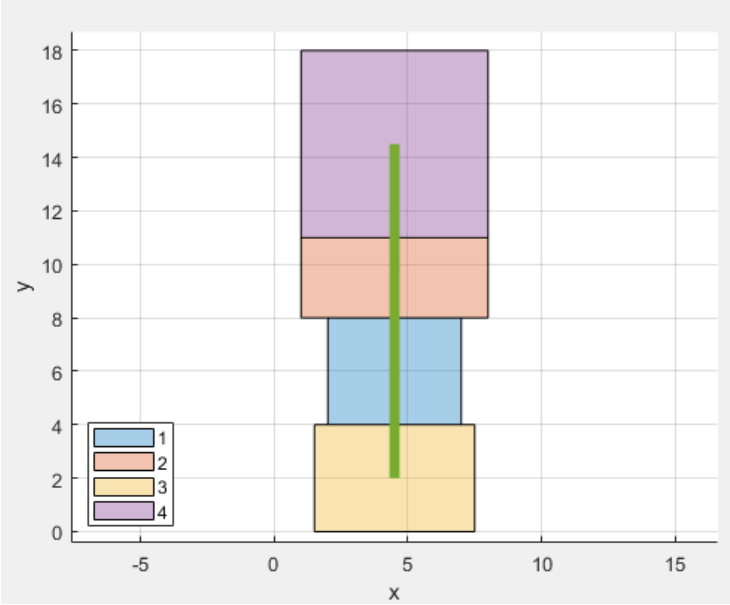
Min $\sum_i \sum_j (R_{ij} + B_{ij})$

Sensitivity: Internal

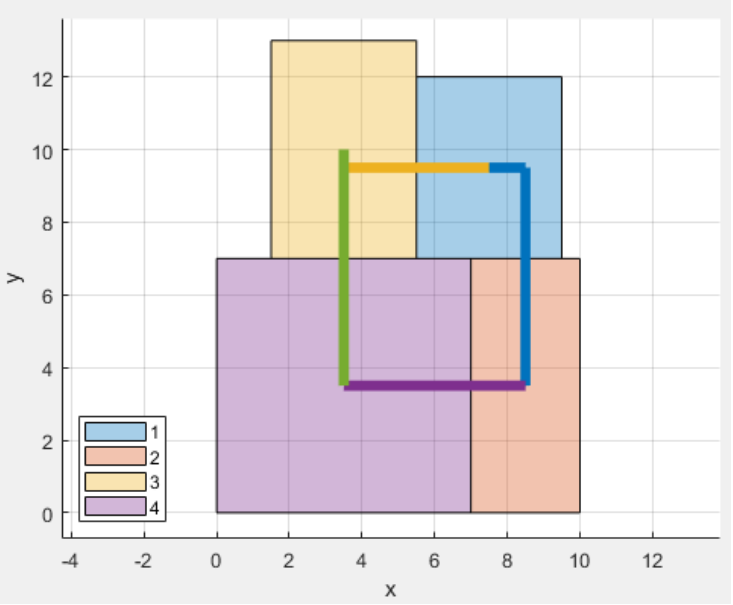
Results for different floor spaces



$W = 20, \quad H = 20$



$W = 8, \quad H = 20$

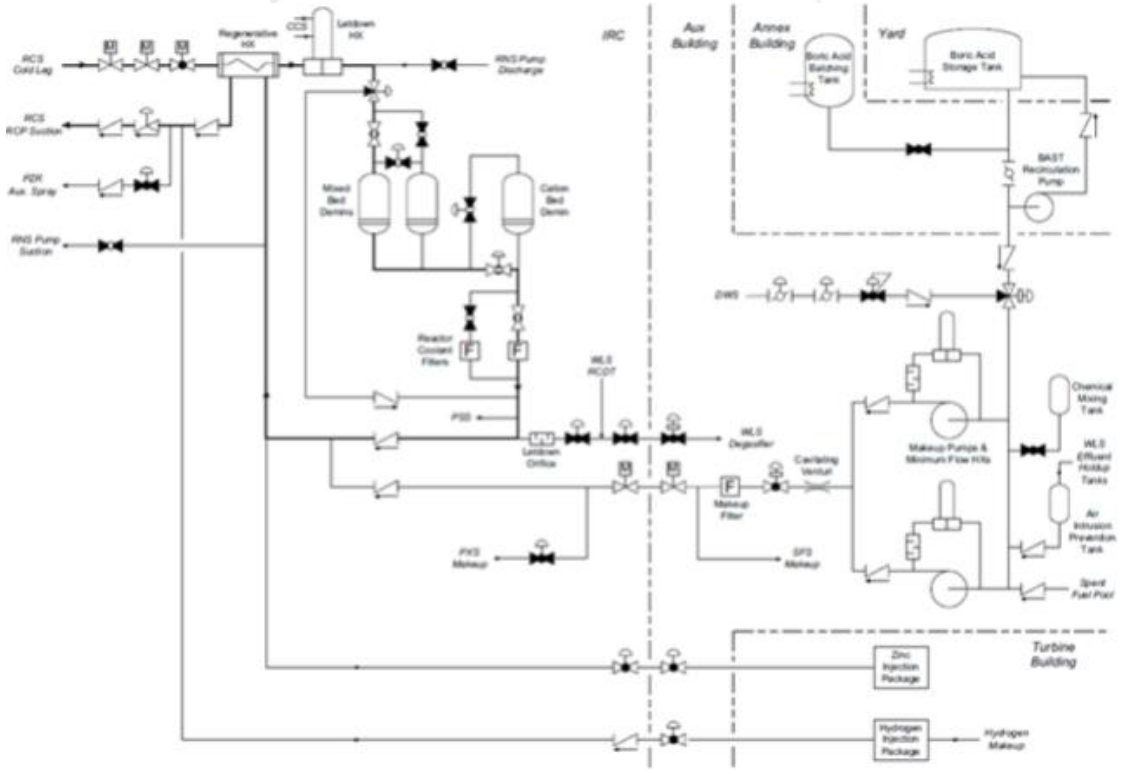


$W = 10, \quad H = 15$

Sensitivity: Internal

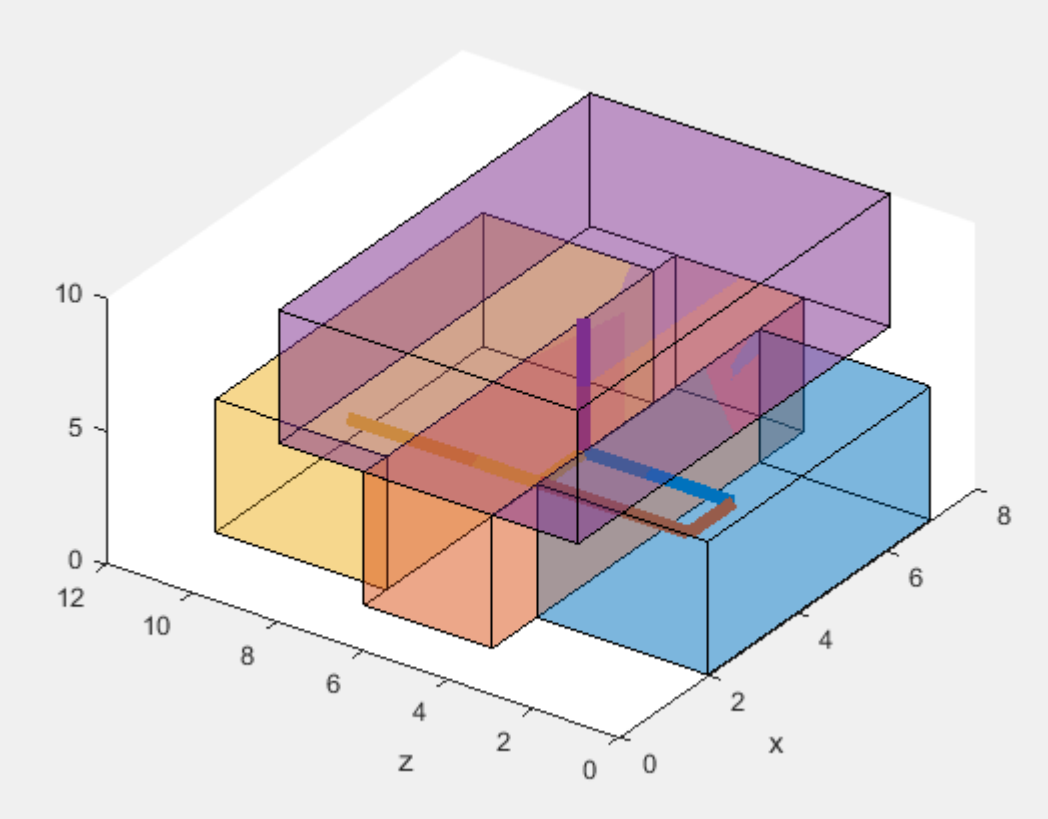
Problem

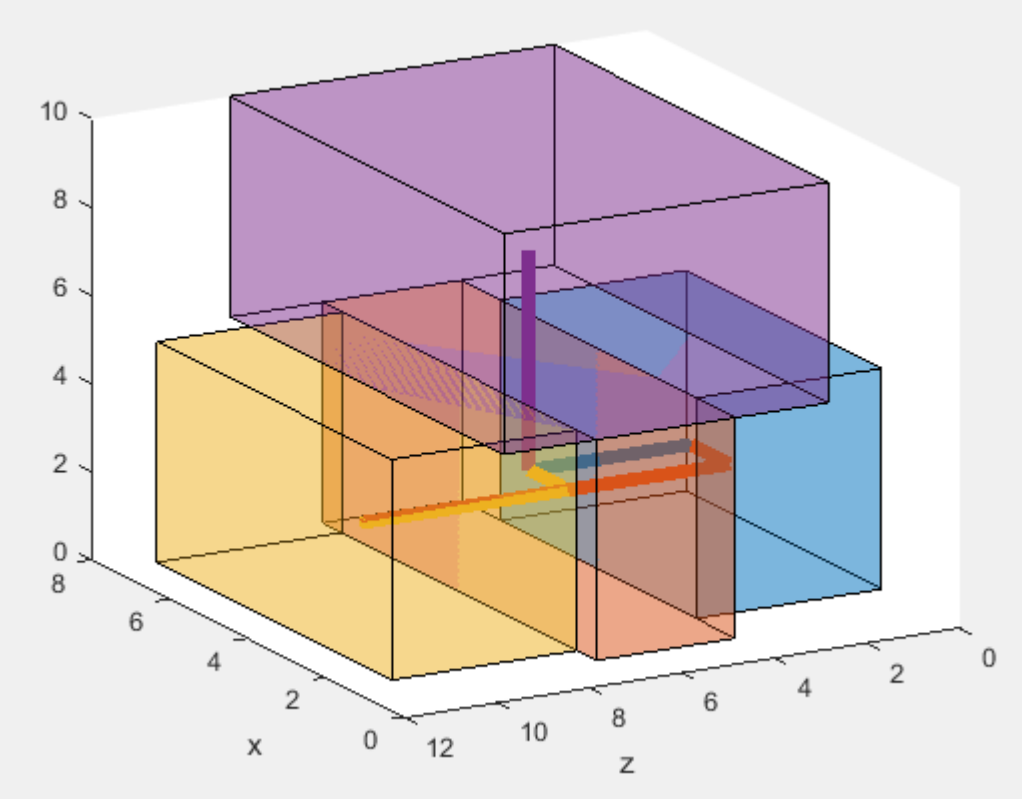




Sensitivity: Internal

Example Solution





This is a simplified plot, the boxes contain enough space to include the piping.
It also requires structures to support items off the floor!

Sensitivity: Internal

Returning to our simple packing problem

minimise A^*

subject to:

For all i (every table)

$$x_i + (1 - r_i)w_i + r_i h_i \leq A^*$$
$$y_i + (1 - r_i)h_i + r_i w_i \leq A^*$$

MiniMax Constraint

For all $i, j,\ i < j$ (all pairs, don't check twice)

$$x_i + (1 - r_i)w_i + r_i h_i \leq x_j + M(x_{ij} + y_{ij})$$
$$x_j + (1 - r_j)w_j + r_j h_j \leq x_i + M(1 - x_{ij} + y_{ij})$$
$$y_i + (1 - r_i)h_i + r_i w_i \leq y_j + M(1 + x_{ij} - y_{ij})$$
$$y_j + (1 - r_j)h_j + r_j w_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

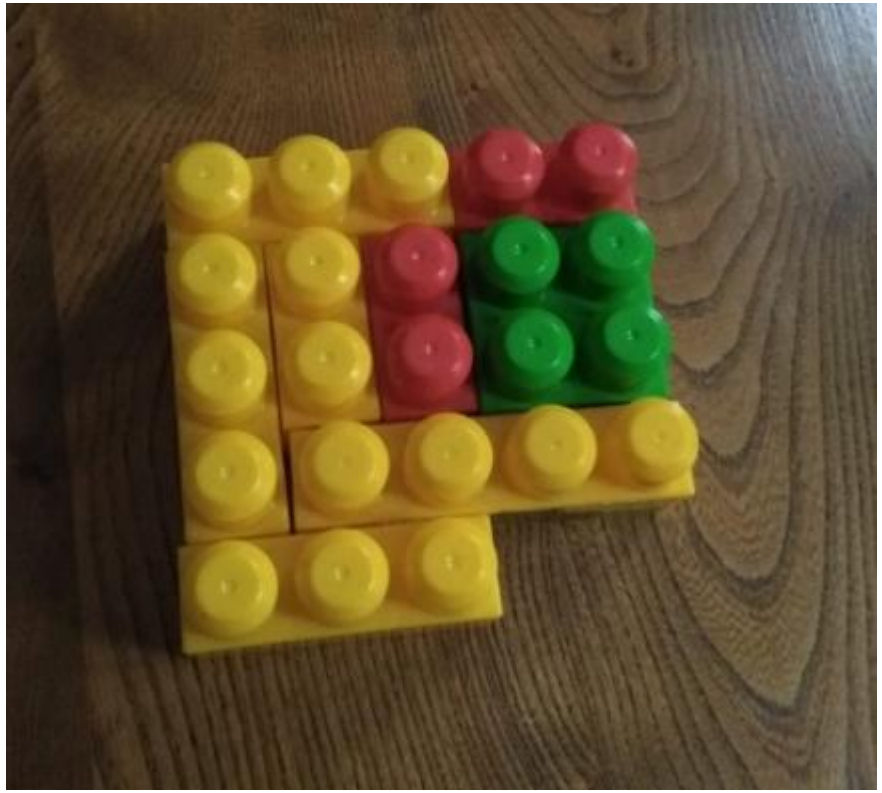
Non overlapping constraints

$$x_{ij}, y_{ij} \in \{0,1\}$$
$$x_i, y_i \geq 0$$

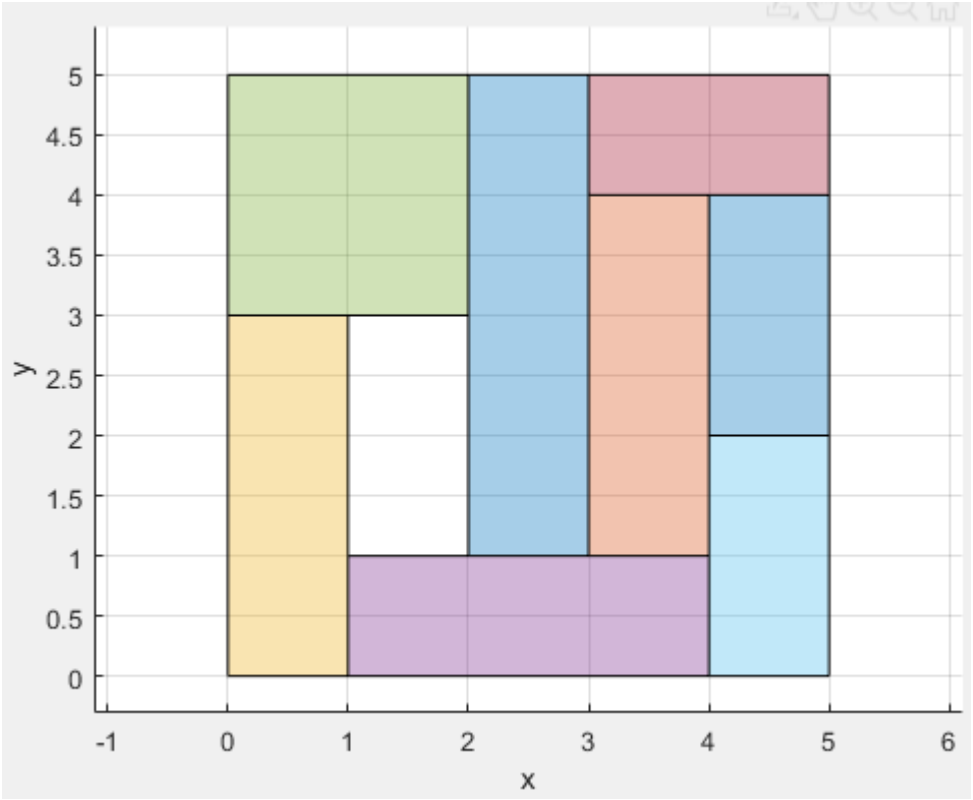
Variable constraints

Sensitivity: Internal

Mums Know Best!



- Approximately 20 seconds
- Approximately 4 solutions explored



- Approximately 10 minutes!
- 4,936,427 solutions explored
- Note that the optimal was actually found in 0.6 seconds, but the program doesn't know it's optimal

Sensitivity: Internal

Why so inefficient?

Binary variables.

x_{ij}, y_{ij} exist for every pair of items

$$2 \cdot \binom{n}{2}$$

Each can have a value of $\{0,1\}$

$$2^{2\cdot\binom{n}{2}}$$

For 7 items that is…

$$2^{2\cdot\binom{7}{2}} \approx 4.4 \times 10^{12}$$

For 10 items to explore every solution every millisecond this would take you

39228790500000000 years!

Sensitivity: Internal

Conclusion

How does Santa pack his sleigh?



NP-Hard Optimisation!

Or By Magic!



Sensitivity: Internal